

CSE-350 Quiz-04

Square Wave Generator, NMOS and CMOS

Set-A

Total – 20 marks

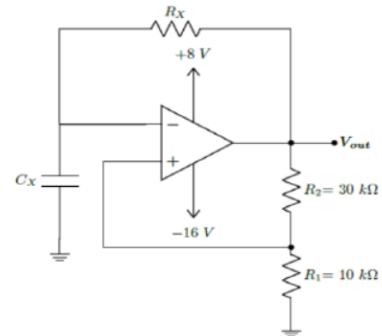
Time – 35 minutes

Name -	Id -	Section -
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Problem-1

10 marks

a) For the given circuit determine the duty cycle	7
b) Plot the input and output on the same graph with proper labeling.	3



Soln: (a) $V_L = -16V$, $V_H = 8V$, $R_1 = 10k$, $R_2 = 30k$

$$V_{UT} = \frac{R_1}{R_1 + R_2} V_H = \frac{10}{10 + 30} \times 8 = 2V$$

$$V_{LT} = \frac{R_1}{R_1 + R_2} V_L = \frac{10}{10 + 30} \times (-16) = -4V$$

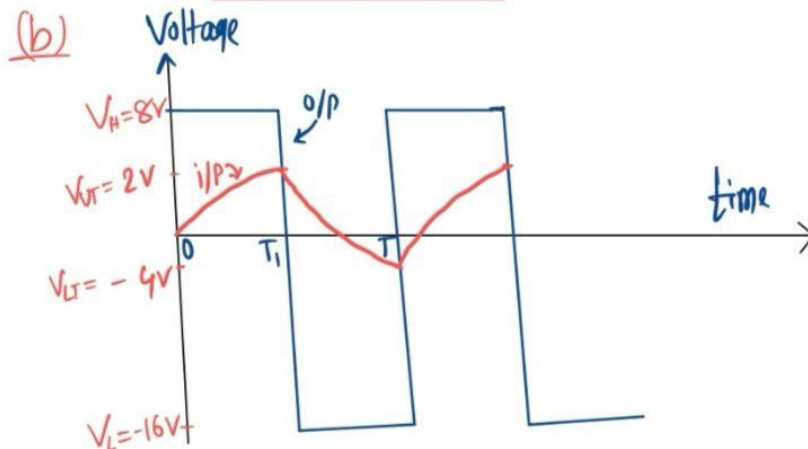
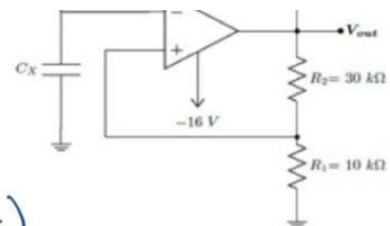
$$T_1 = R_X C_X \ln \left(\frac{V_H - V_{LT}}{V_H - V_{UT}} \right) = R_X C_X \ln \left(\frac{8 - (-4)}{8 - 2} \right) = R_X C_X (0.69)$$

$$T_2 = R_X C_X \ln \left(\frac{V_L - V_{UT}}{V_L - V_{LT}} \right) = R_X C_X \ln \left(\frac{-16 - 2}{-16 - (-4)} \right) = R_X C_X (0.405)$$

$$\text{Duty cycle, } D = \frac{T_1}{T_1 + T_2} \times 100\%$$

$$= \frac{R_X C_X \times 0.69}{R_X C_X (0.69 + 0.405)} \times 100\%$$

$$D = 63.01\%$$



Question 2

A Non-inverting Schmitt trigger circuit having a center voltage of 2V and Hysteresis width of 2V. Assume $V_H = +4V$ and $V_L = -4V$. Draw the circuit and transfer curve with proper labeling.

(a)	Design the circuit with all component values.	2.5
(b)	Draw V_{out} vs. V_{in} graph (VTC) or Transfer curve with proper labeling.	2.5

$$\Rightarrow \text{given} \rightarrow V_{CTR} = V_S = 2V, V_{HYS} = 2V, V_H = 4V, V_L = -4V$$

$$V_{UT} = V_{CTR} + \frac{V_{HYS}}{2} \quad V_{LT} = V_{CTR} - \frac{V_{HYS}}{2}$$

$$= 2 + 1 = 3V \quad = 2 - 1 = 1V$$

$$V_{CTR} = V_S = \frac{R_1 + R_2}{R_2} V_{REF} = 2V$$

$$V_{UT} = -\frac{R_1}{R_2} V_L + \frac{R_1 + R_2}{R_2} V_{REF}$$

$$3 = -\frac{R_1}{R_2} (-4) + 2$$

$$1 = \frac{R_1}{R_2} (4)$$

$$\therefore R_2 = 4R_1$$

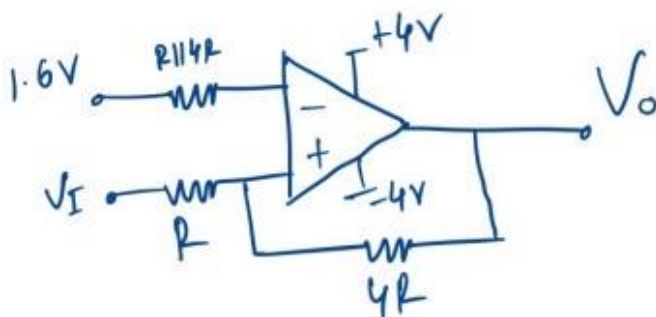
Let $\rightarrow R_1 = R, R_2 = 4R$
(you can choose any value for R)

$$\frac{R_1 + R_2}{R_2} V_{REF} = 2$$

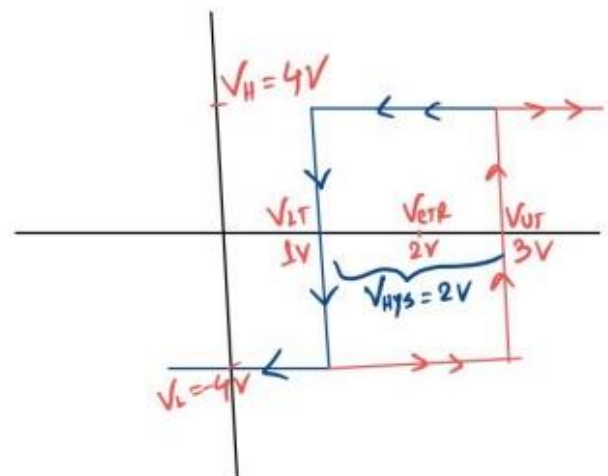
$$\Rightarrow \frac{R + 4R}{4R} V_{REF} = 2$$

$$\Rightarrow \frac{5}{4} V_{REF} = 2$$

$$\therefore V_{REF} = \frac{8}{5} = 1.6V$$



$\therefore$ Non-Inv. Schmitt trigger with applied voltage



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Set-B

Total – 20 marks

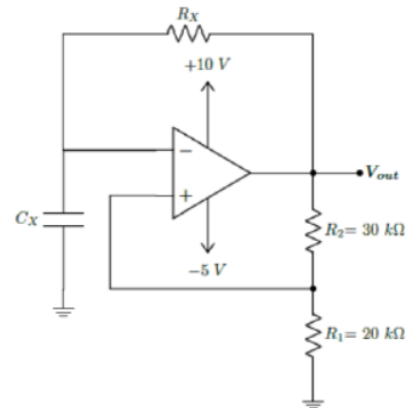
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Problem-1

10 marks

a) For the given circuit determine the duty cycle	7
b) Plot the input and output on the same graph with proper labeling.	3



Same as set-A, just value changed

Question 2

A Non-inverting Schmitt trigger circuit having a center voltage of 2V and Hysteresis width of 2V. Assume $V_H = +4V$ and $V_L = -4V$. Draw the circuit and transfer curve with proper labeling.

(a)	Design the circuit with all component values.	2.5
(b)	Draw V_{out} vs. V_{in} graph (VTC) or Transfer curve with proper labeling.	2.5

$$\Rightarrow \text{given} \rightarrow V_{CTR} = V_S = 2V, V_{HYS} = 2V, V_H = 4V, V_L = -4V$$

$$V_{UT} = V_{CTR} + \frac{V_{HYS}}{2} \quad V_{LT} = V_{CTR} - \frac{V_{HYS}}{2}$$

$$= 2 + 1 = 3V \quad = 2 - 1 = 1V$$

$$V_{CTR} = V_S = \frac{R_1 + R_2}{R_2} V_{REF} = 2V$$

$$V_{UT} = -\frac{R_1}{R_2} V_L + \frac{R_1 + R_2}{R_2} V_{REF}$$

$$3 = -\frac{R_1}{R_2} (-4) + 2$$

$$1 = \frac{R_1}{R_2} (4)$$

$$\therefore R_2 = 4R_1$$

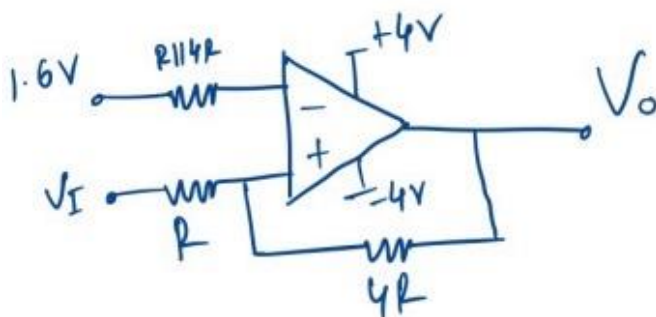
Let $\rightarrow R_1 = R, R_2 = 4R$
(you can choose any value for R)

$$\frac{R_1 + R_2}{R_2} V_{REF} = 2$$

$$\Rightarrow \frac{R + 4R}{4R} V_{REF} = 2$$

$$\Rightarrow \frac{5}{4} V_{REF} = 2$$

$$\therefore V_{REF} = \frac{8}{5} = 1.6V$$



$\therefore$ Non-Inv. Schmitt trigger
with applied voltage

